MIL-S-19500/232B
24 June 1970
SUPERSEDING
MIL-S-19500/232A
28 March 1963

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, MIXER

TYPES 1N21WE, 1N21WEM, AND 1N21WEMR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon semiconductor diode (with removable adapter) types: 1N21WE (forward polarity); 1N21WEM (matched forward pair); and 1N21WEMR (matched forward and reverse); for use as a mixer in S-band equipment. Ratings for the 1N21WE are applicable to the types having additional suffixes (see 3.5.1).

1.2 Ratings and characteristics:

	$\bar{F}_o$	1/G	L	N	SWR	Burnout
	dB	ohms	dB	ratio	ratio	erg
Minimum	---	350	---	---	---	---
Maximum	7.0	450	5.5	1.5	1.3	2.0

OPERATING AMBIENT TEMPERATURE: -65° to +150°C

STORAGE TEMPERATURE: -65° to +150°C

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification For.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

DRAWINGS

Defense Electronics Supply Center

D64100 - Test Holder for Microwave Diode Type 1N21.
C66058 - Burnout Tester for Microwave Diodes.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements for semiconductor diodes shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500, and as follows:

1/G - - - - - - - - - - Reciprocal of conductance (IF impedance).
SWR- - - - - - - - - - Standing wave ratio.

3.3 Design, construction, and physical dimensions. The semiconductor diode shall be of symmetrical construction and shall consist of a cylindrical body having pins at both ends. The body outline with removable base adapter shall conform to figure 1.

3.3.1 Base adapter. The base adapter shall be of the design, construction, and physical dimensions shown in figure 2.

3.3.2 Plating. The diode and base adapter shall be plated as specified in figures 1 and 2.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The marking shall be as specified in MIL-S-19500 except that the manufacturers identification and country of origin may be omitted.

3.5.1 Matched diodes. The "M" suffix for matched diodes shall be omitted in the type designation on each device. Matched diodes meeting the requirements of this specification shall be packaged with a statement to that effect.

3.6 Burnout single pulse. At the end of manufacturing processes and prior to selecting samples for testing, all diodes shall be subjected to 100-percent burnout by single-pulse test, only once, and this test shall be performed in accordance with method 4146, MIL-STD-750, using Drawing C66058, E = 125 Vdc (minimum) and with the adapter affixed for forward-direction operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Test conditions. Unless otherwise specified herein, the test conditions (when applicable) shall be as follows:

P = 0.5 mW ±5%

f = 3,060 ±5 MHz

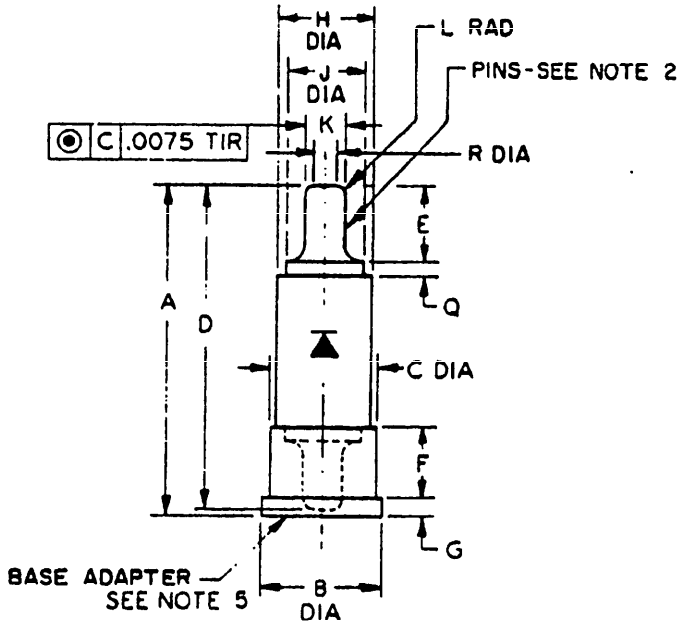
$\overline{F_i} \geq$ 1.5 dB

Z_L = 400 ±10 ohms; ±j0 ohms

R_L = 100 ±1 ohms

Drawing D64100

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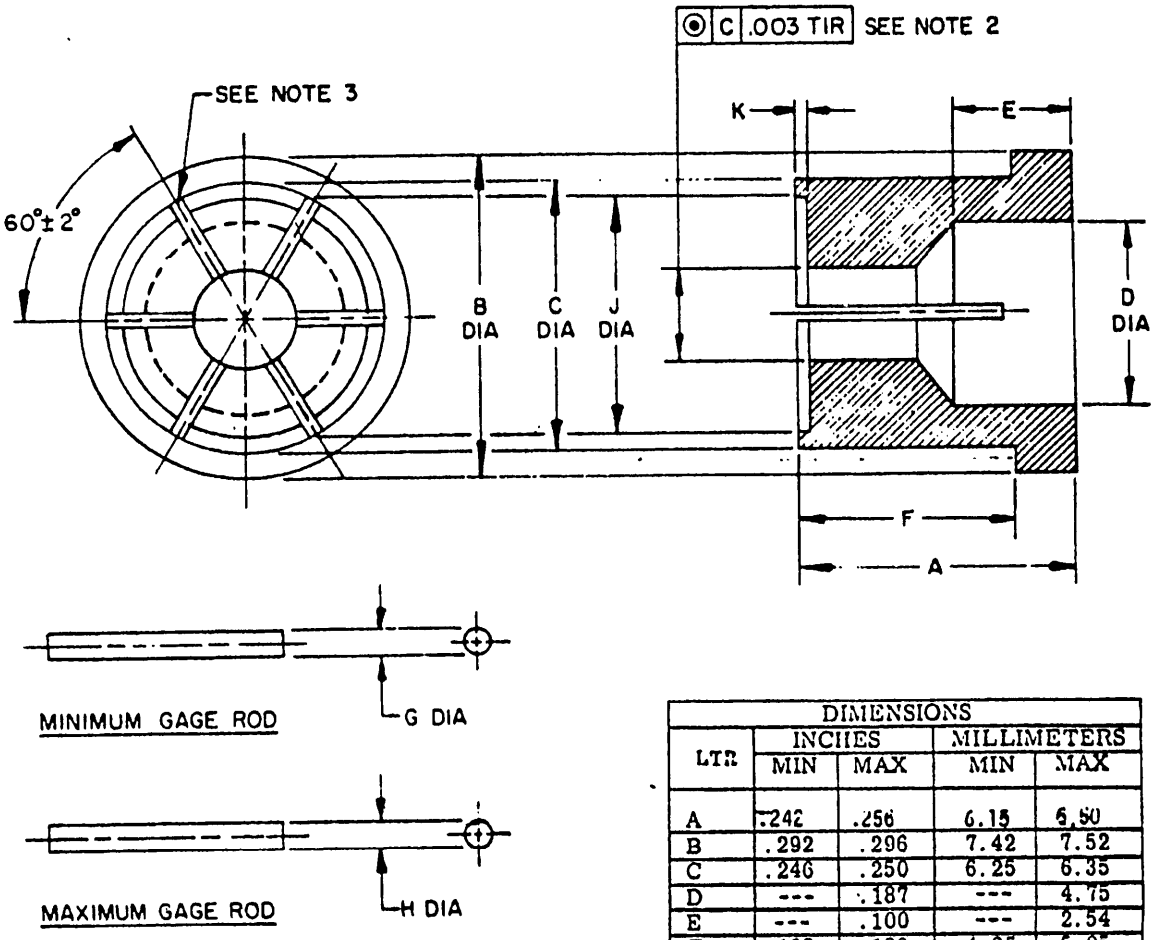


DIMENSIONS					
LTR	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	.800	.840	20.32	21.34	
B	.292	.296	7.42	7.52	
C	.246	.250	6.25	6.35	
D	.766	.792	19.46	20.12	
E	.180	.190	4.57	4.83	6
F	.195	.199	4.95	5.05	
G	.047	.057	1.19	1.45	
H	.222	.240	5.64	6.10	
J	.195	.215	4.95	5.46	6
K	.092	.094	2.34	2.39	2, 5, 6
L	.030	.046	.76	1.17	6
Q	.020	.030	.51	.76	6
R	---	.030	---	.76	6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. The ends of both pins shall be smooth with no burrs or sharp edges. Within the diameter R a recess is allowed on the ends of the pins.
3. Metal parts shall be gold plated 10 msi minimum.
4. The polarity shall be indicated by an arrow, with the arrow pointing in the direction of easier current flow.
5. Removable base adapter shall establish the forward or reverse polarity.
6. Applies to both pins.

FIGURE 1. Semiconductor device, diode, types 1N21WE, 1N21WEM, and 1N21WEMR.

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DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.242	.256	6.15	6.50
B	.292	.296	7.42	7.52
C	.246	.250	6.25	6.35
D	---	.187	---	4.75
E	---	.100	---	2.54
F	.195	.199	4.95	5.05
G	.0915	.0920	2.32	2.34
H	.0940	.0945	2.39	2.40
J	.216	.221	5.49	5.61
K	.031	.036	.79	.91

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. This diameter should be .100 (2.54 mm) max diameter before closing jaws. This diameter shall then be sufficiently closed, and the adapter so tempered that it will fit on the minimum gage rod and also on the maximum gage rod, in each case with a snug fit (for hand assembly).
3. Each slot shall be .013 (.330 mm) min, .017 (.432 mm) max wide by .185 (4.70 mm) deep before closing. The 6 slots are equally spaced.
4. The material for the adapter shall be beryllium copper or half-hard brass per QQ-B-626, composition 22.
5. Metal parts shall be gold plated 10 msi minimum.
6. All burrs and sharp edges shall be removed.

FIGURE 2. Removable base adapter, included as a part of types 1N21WE, 1N21WEM, and 1N21WEMR.

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4.2.1 Holder. The diode holder specified in Drawing D64100 (or equivalent) shall be used for all electrical test measurements.

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III, except the matched-pair requirements in subgroup 3, table I.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in groups A, B, and C.

4.4.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.4.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.4.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III.

4.5 Method of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.5.1 High-temperature operation. The semiconductor diode shall be placed in the mixer holder. The ambient temperature of the diode, with test conditions specified for the overall noise figure, shall be raised to +150°C and maintained at this temperature until equilibrium is reached. The $\overline{F}_O$ shall then be determined and shall be less than 12 dB. The temperature shall then be returned to 25 ±3°C at which time $\overline{F}_O$ shall be less than the specified limit.

4.5.2 Matched pair, "M" suffix (forward polarity). The matched forward pair (M suffix) shall consist of two diodes, tested to requirements of subgroup 3, table I, having the anode connected to the adapter. (See 3.5.1.)

4.5.3 Matched pair, "MR" suffix (forward and reverse polarity). The matched forward and reverse pair (MR suffix) shall consist of two diodes, tested to requirements of subgroup 3, table I: one diode having the anode connected to the adapter (forward polarity) and the second diode having the cathode connected to the adapter (reverse polarity). (See 3.5.1.)

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			7				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Standing wave ratio (see 4.2)	4136	Test method A		SWR	---	1.3	ratio
Overall (average) noise figure (see 4.2)	4126	Test method A		$\overline{F}_O$	---	7.0	dB
IF impedance (see 4.2)	4116	f = 60-2,000 Hz		1/G	350	450	ohms

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TABLE I. Group A inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 3			5				
Matched pair requirements: (see 4.5.2 and 4.5.3)							
Conversion loss unbalance	4101			ΔL	---	0.3	dB
IF impedance unbalance	4116			Δ(1/G)	---	25	ohms

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1							
Physical dimensions	2066	(See figures 1 and 2.)					
		Dim. C, J, K, and note 2 of Figure 1	7	---	---	---	---
		Dim. A, B, D, E, F, G, H, L, and Q of Figure 1. (Note 4 of figure 2 is for qualification only.)	20	---	---	---	---
Subgroup 2			10				
Thermal shock (temperature cycling)	1051	Test condition F		---	---	---	---
Terminal strength:							
Tension	2036	Test condition A; 1 lb; t = 30 sec		---	---	---	---
Torque	2036	Test condition D1; t = 30 sec		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points:							
Overall (average) noise figure (see 4.2)	4126	Test method A		$\bar{F}_o$	---	8.0	dB
Subgroup 3			10				
Burnout by repetitive pulsing (forward polarity)	4141	e_o = 15 v; R_g = 50 ohms, 60,000 pulses (min); t_p = 0.05 μsec; PRF optional		---	---	---	---
End points: (Same as subgroup 2)							

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 4			10				
Shock	2016	Nonoperating; 500 G; t = 1.0 msec; 5 blows in each orientation: X_1, Y_1, and Y_2		---	---	---	---
Vibration, variable frequency	2056	Nonoperating; 15 G, 50 to 2,000 Hz		---	---	---	---
Constant acceleration	2006	Nonoperating; 20,000 G, X_1, Y_1, and Y_2 orientations		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 5			10				
High temperature operation: (see 4.5.1)	---	T_A = 150°C		---	---	---	---
Overall (average) noise figure (see 4.2)	4126	Test method A		$\overline{F}_O$	---	8.0	dB
Subgroup 6			λ=15				
High temperature life (nonoperating)	1031	T_A = 150°C		---	---	---	---
End points: (Same as subgroup 2)		–					

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Conversion loss (see 4.2)	4101			L	---	5.5	dB
Output noise ratio (see 4.2)	4121	R_e = 400 ohms		N	---	1.5	ratio

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Substitution information. Type 1N21WE (with its removable adapter) may be substituted for the corresponding commercial types: 1N21E, 1N21ER, 1N21EM, and 1N21EMR.

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Navy - EC
Air Force - 17

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Navy - SH
Air Force - 11, 70, 85
DSA - ES

User activities:
Army - SM
Navy - AS, OS, MC, CG
Air Force - 13, 15, 19, 80

Preparing activity:
Navy - EC

Agent:
DSA - ES

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